



BIO-MEDICAL INFORMATICS

"Mario Stefanelli"



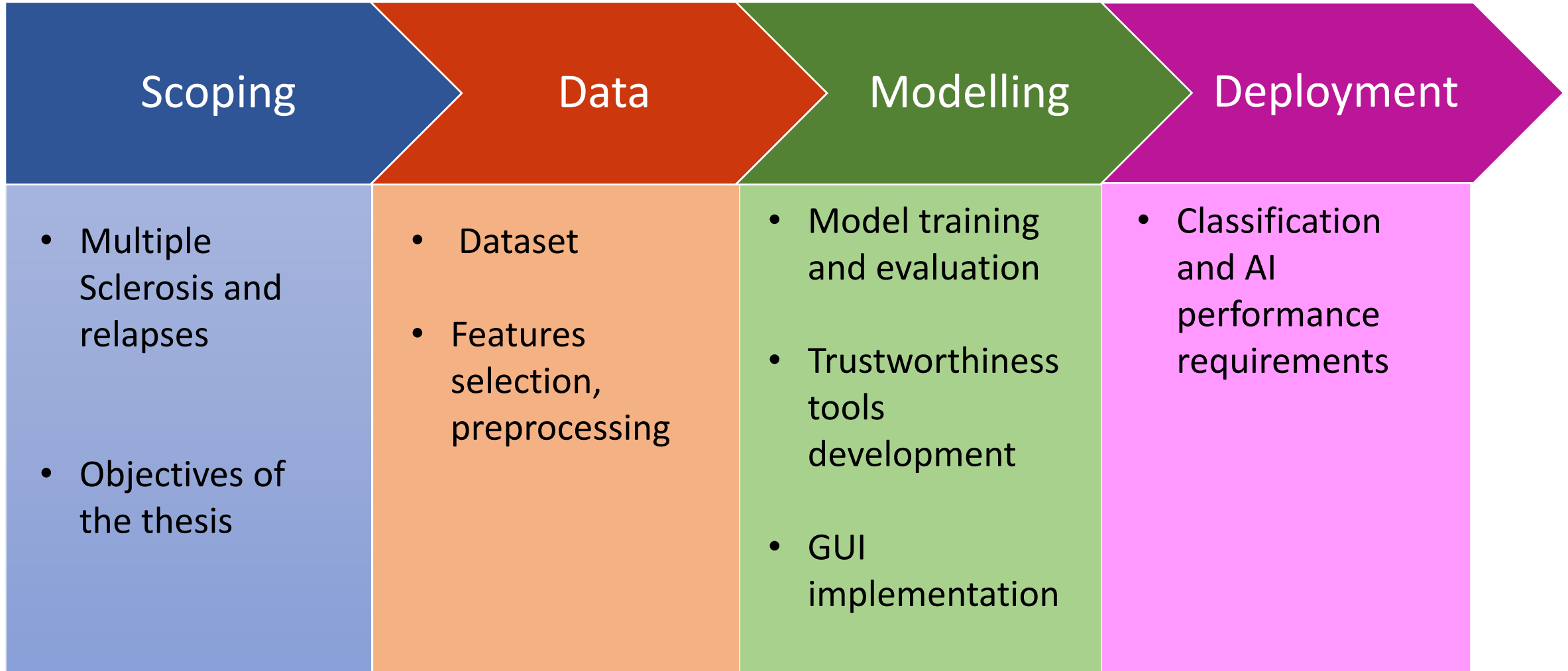
A new AI-based Medical Device to predict Multiple Sclerosis worsening: design, development and trustworthiness in clinical decision support

Bachelor in Artificial Intelligence

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A.Y. 2023/2024

Thesis Outline



Scoping

Data

Modelling

Deployment

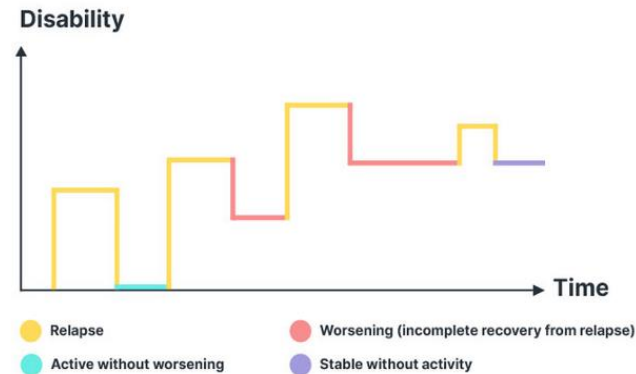
Introduction

Multiple Sclerosis

- Chronic disease of the central nervous system
- Wide range of symptoms including motor, sensory and cognitive problems
- The majority of people diagnosed suffer from sudden RELAPSES

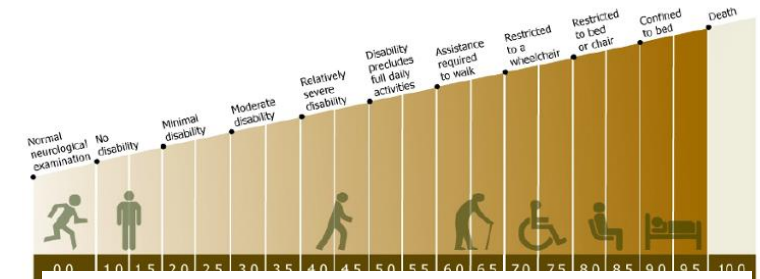
Acute episodes of illness

- The majority of people diagnosed suffer from sudden **RELAPSES**



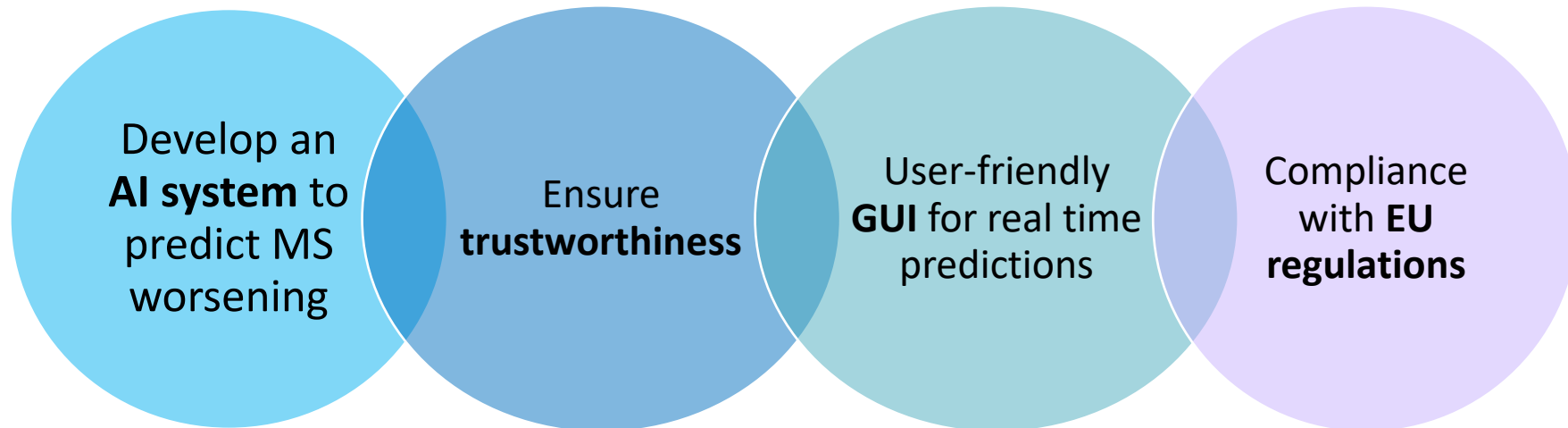
Expanded Disability Status Scale

- How do we define **WORSENING**?
- ↓
- EDSS



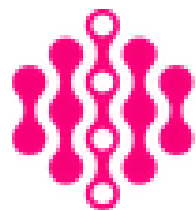


Objectives of the thesis





The dataset



Brainteaser

Bringing Artificial Intelligence
home for a better care of
amyotrophic lateral sclerosis
and multiple sclerosis

LifeSTech 

800 ANNI



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



 Ciências
ULisboa



 Hospital General Universita
Gregorio Marañón

 InSilicoTrials



TRAIN		TEST	
Negative	Positive	Negative	Positive
219	35	60	11



Data preprocessing

Feature selection:

- Sex
- Residence classification
- MS in pediatric age
- Spinal cord symptoms
- Brainstem symptoms
- Eye symptoms
- Supratentorial symptoms
- Other symptoms
- Latent class (LCMM)

Categorical features

OneHotEncoder

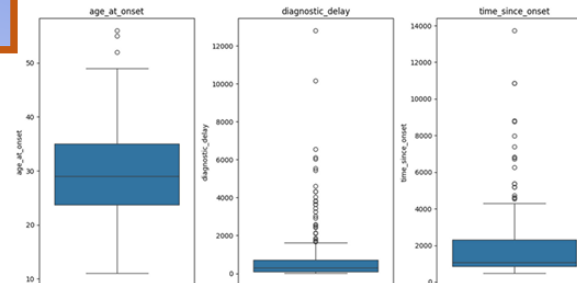
- Time since onset
- Age at onset
- Diagnostic delay

Numerical features

RobustScaler

Class imbalance

SMOTE

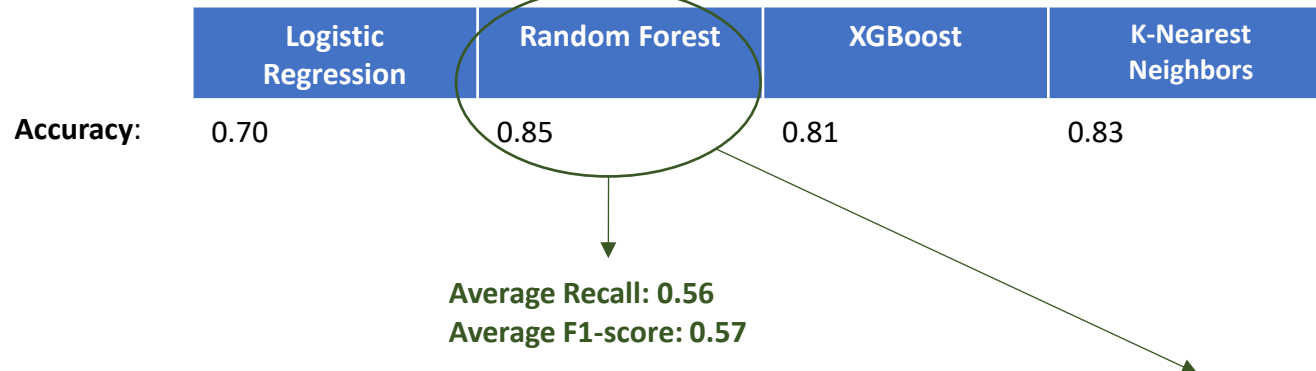




Model training and evaluation

- Comparison of various models on the validation set:

- 300 decision trees
- Maximum depth of 20



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Results of the **Random Forest** on the test set:

Accuracy	Average Recall	Average F1-score
0.8	0.68	0.66



User interface

with **Streamlit**

- User input
- 1 Patient categorical features
 - 2 Patient numerical features
 - 3 Press the **“Compute prediction”** button to visualize the output

sex
male

residence_classification
Cities

ms_in_pediatric_age
False

spinal_cord_symptom
False

brainstem_symptom
False

eye_symptom
False

supratentorial_symptom
True

other_symptoms
sensory

latent_class_icmm
1

1

2

3

4

5

age_at_onset
49

diagnostic_delay
431

time_since_onset
1993

Compute prediction

Result:

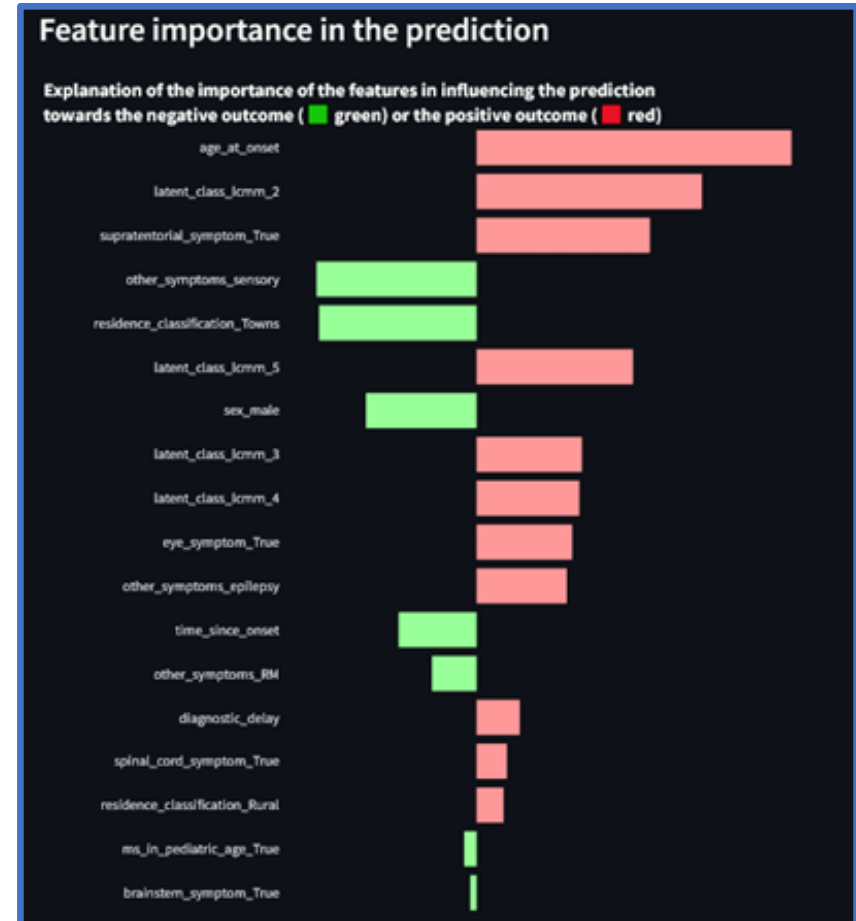
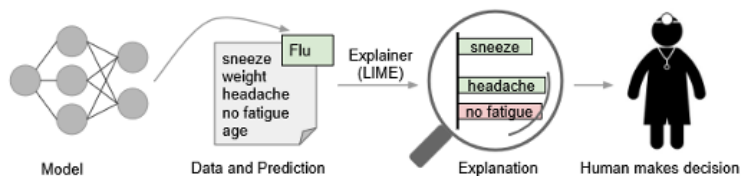
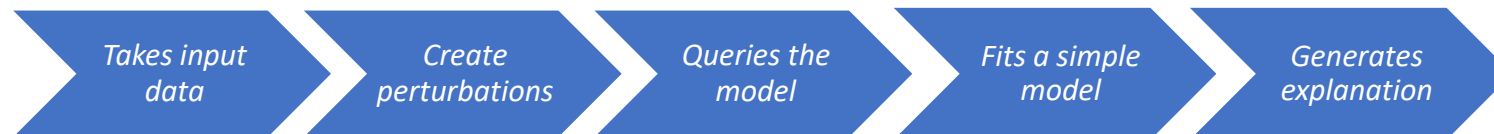
Predicted outcome: Positive: the patient will exceed the EDSS ≥ 3 within the next year

Confidence about the prediction: 74.00%



LIME

- Ensuring EXPLAINABILITY
- Local Interpretable Model-Agnostic Explanations
- “What part of the input led the model to make this specific prediction for this specific patient?”

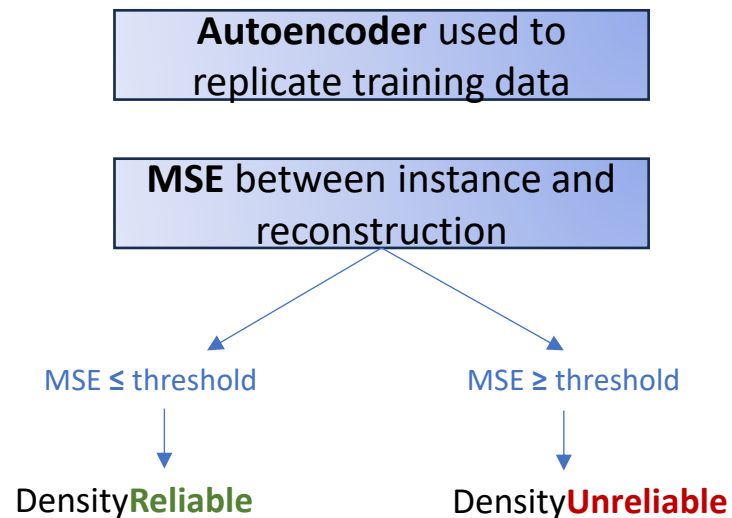




RELIABILITY of the model on each single patient using RelAI

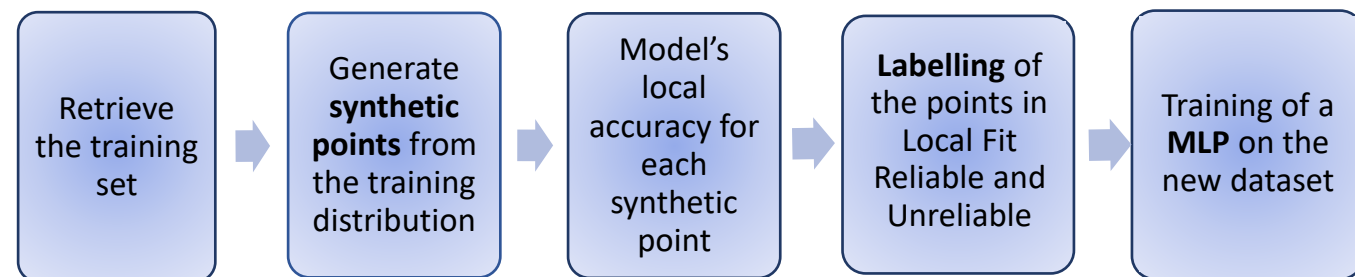
DENSITY PRINCIPLE

- It asks whether a sample is similar to the training set



LOCAL FIT PRINCIPLE

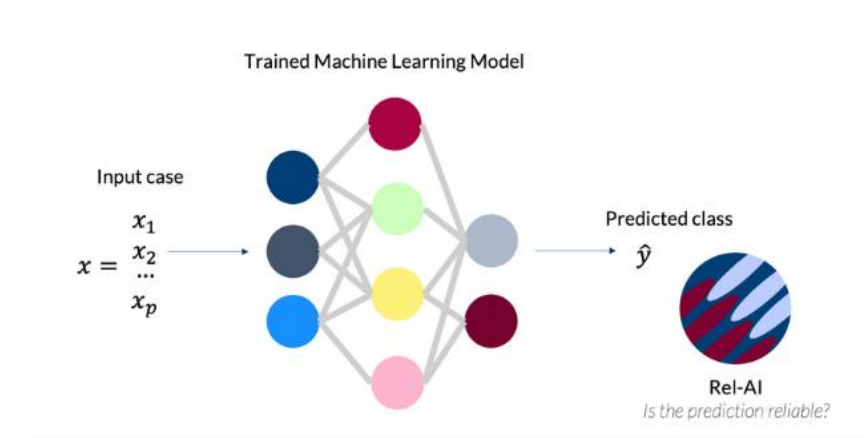
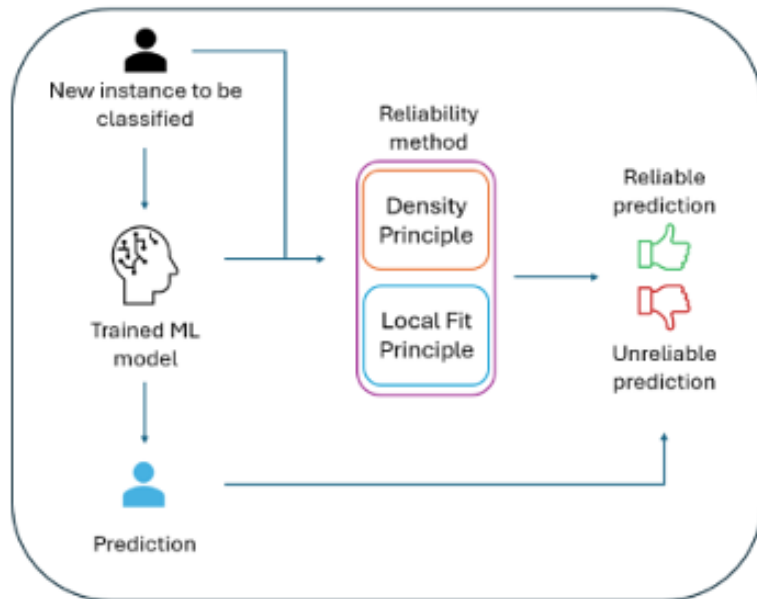
- It asks whether the model was well performing on cases similar to the new case to be classified





RELIABILITY of the model on each single patient using RelAI

- Total reliability is satisfied if both principles are respected



Reliability metrics of the prediction

Density Principle : The patients data closely resembles cases the model was trained on

Reconstruction error (MSE): 0.00693

Local Fit Principle : The model has consistently performed well on similar cases

Total reliability : The prediction is considered reliable



Classification and AI performance requirements

Medical device

According to the EU MDR: software used for specific medical purposes

Class IIb

Under Rule 11 for MDR, Annex VIII: monitors parameters that could result in immediate danger

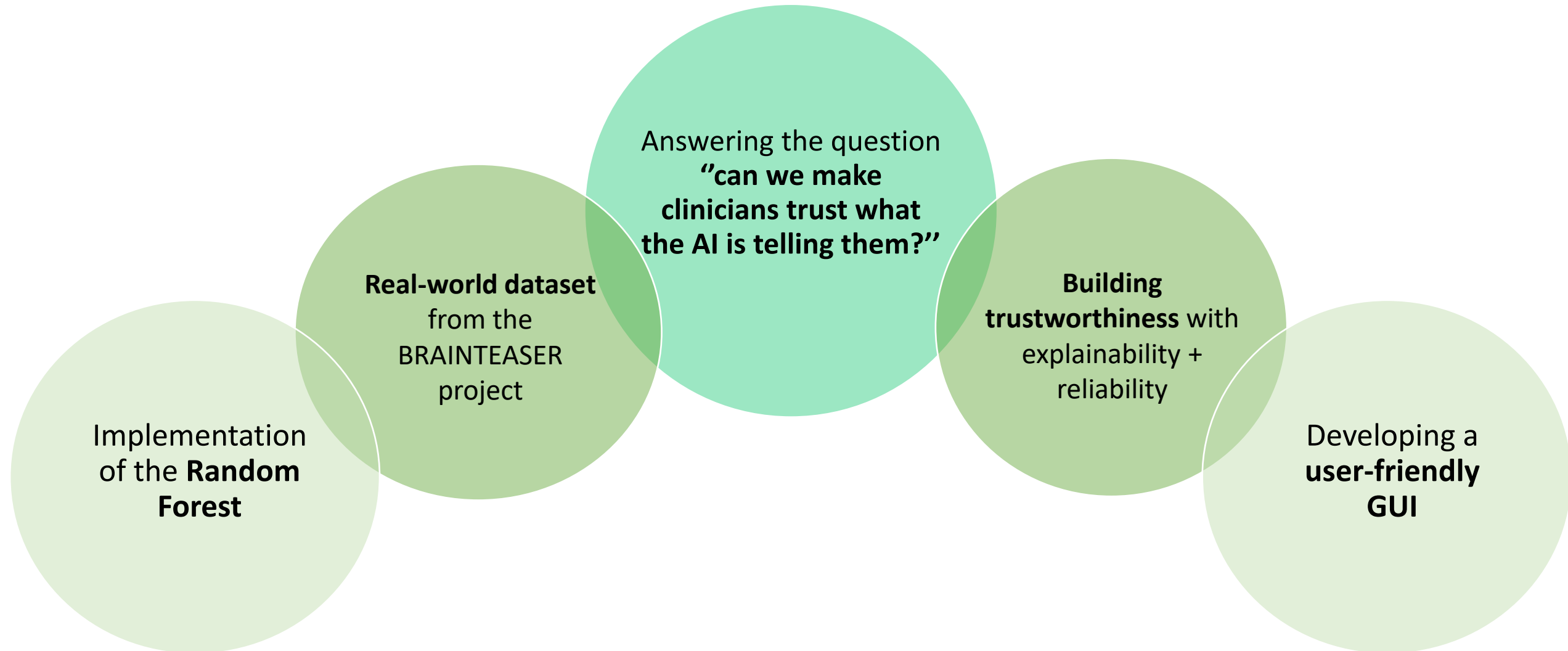
Follows the GDPR

No input of sensitive data, no tracking

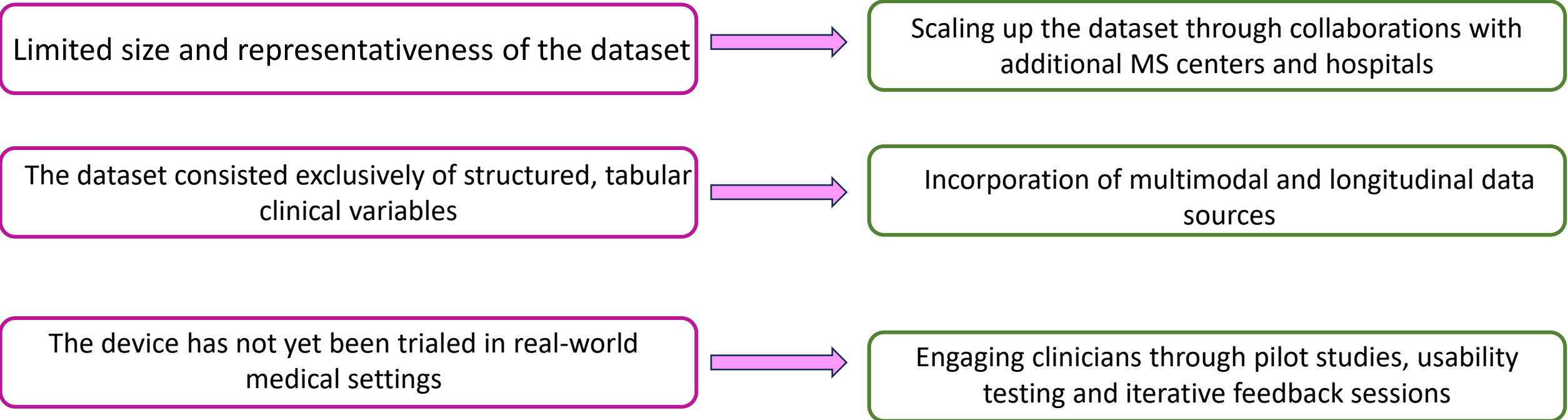
Adheres to MLOps principles

Defining clinical objective, data preparation, developing of model, trustworthiness techniques and GUI, deployment

Results and achievements



Limitations and future work



Thanks for your attention!